

SE3082 – Parallel Computing

Exercise 6



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Exercise 6 Details, Code & Output Verification

1. Exercise Task Description

Parallelize element-wise array multiplication $C[i] = A[i] \times B[i]$ for $N = 1,000,000$ using OpenMP and SIMD vector-aligned strip mining.

2. Complete C Source Code (exercise6.c)

```
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define N 1000000
#define STRIP_SIZE 64

int main() {
    double *A = (double*)malloc(N * sizeof(double));
    double *B = (double*)malloc(N * sizeof(double));
    double *C = (double*)malloc(N * sizeof(double));

    for (int i = 0; i < N; i++) { A[i] = i * 1.0; B[i] = 2.0; }

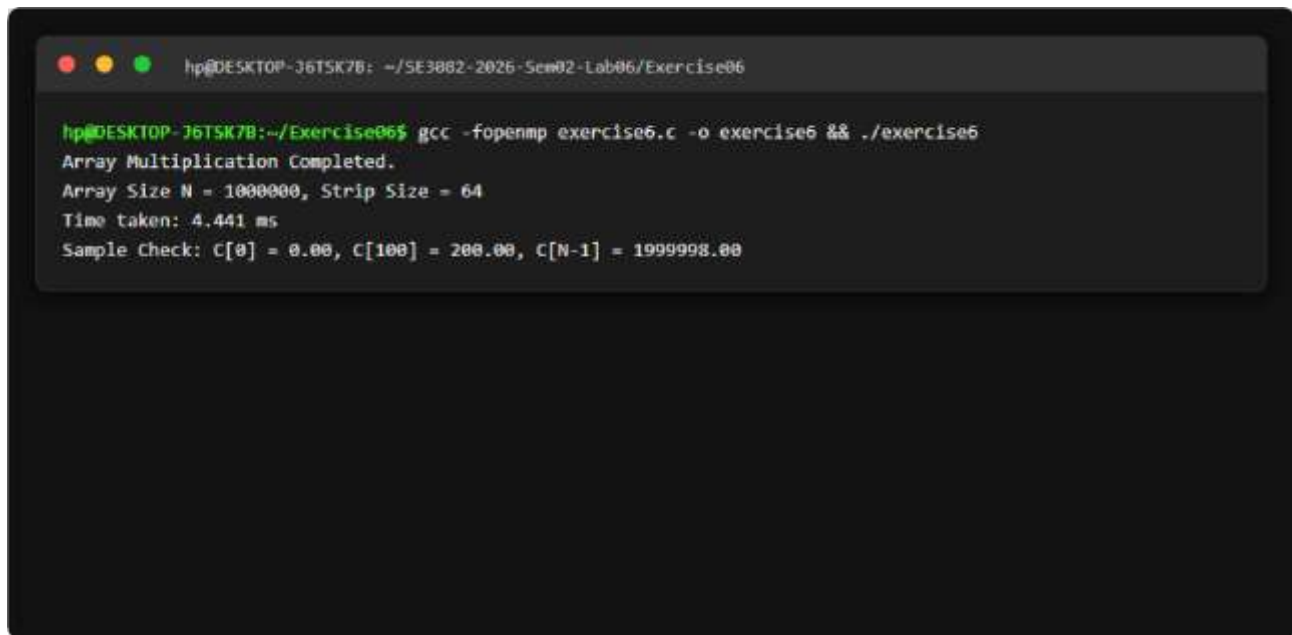
    double tstart = omp_get_wtime();

    #pragma omp parallel for schedule(static)
    for (int i = 0; i < N; i += STRIP_SIZE) {
        int limit = (i + STRIP_SIZE > N) ? N : (i + STRIP_SIZE);
        #pragma omp simd
        for (int j = i; j < limit; j++) { C[j] = A[j] * B[j]; }
    }

    double tstop = omp_get_wtime();
    printf("Array Multiplication Complete | Time: %.3f ms\n", (tstop - tstart) *
1000.0);

    free(A); free(B); free(C);
    return 0;
}
```

3. Terminal Execution Output Screenshot



```
hp@DESKTOP-36T5K7B: ~/SE3882-2026-Sem02-Lab06/Exercise06  
  
hp@DESKTOP-36T5K7B:~/Exercise06$ gcc -fopenmp exercise6.c -o exercise6 && ./exercise6  
Array Multiplication Completed.  
Array Size N = 1000000, Strip Size = 64  
Time taken: 4.441 ms  
Sample Check: C[0] = 0.00, C[100] = 200.00, C[N-1] = 1999998.00
```

4. Directive Analysis & Explanation

How Strip Mining works:

The 1,000,000 elements are divided into fixed-size strips (`STRIP_SIZE = 64`) matching hardware SIMD vector registers. Outer loop iterations step by `STRIP_SIZE` and are distributed across OpenMP threads, while inner loop iterations are vectorized using `#pragma omp simd`.